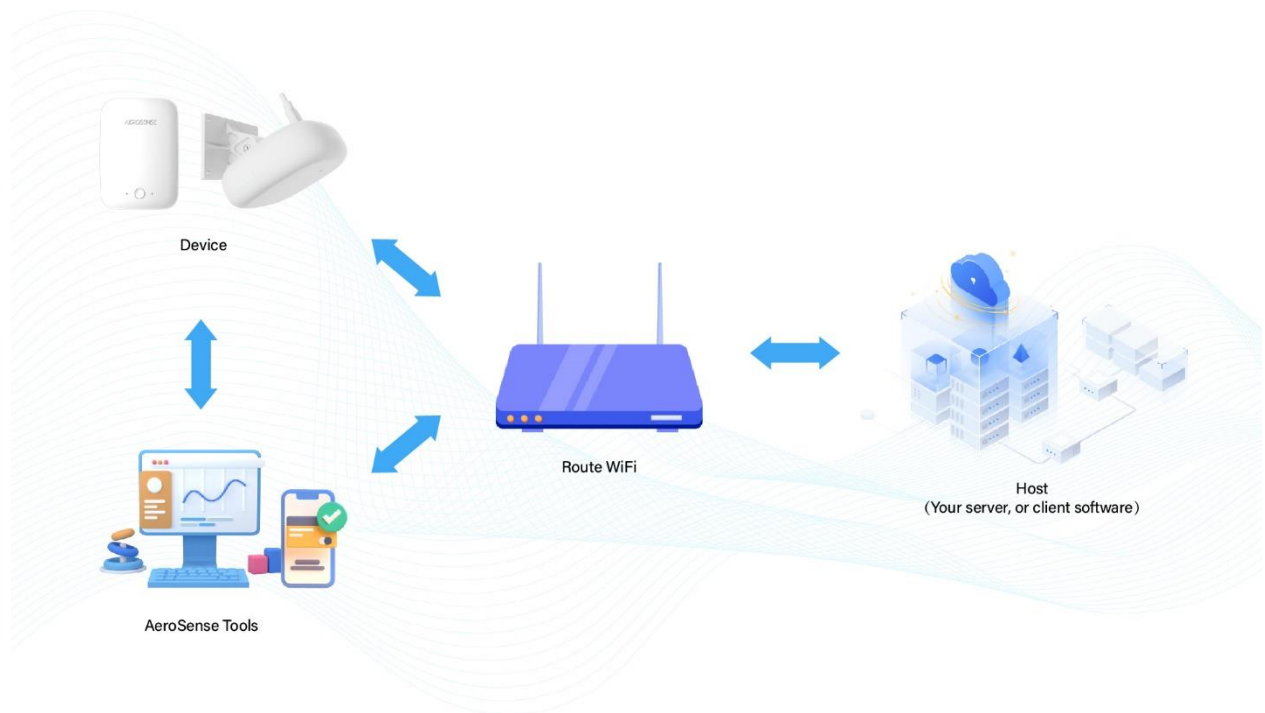


AEROSENSE



Assure and Wavve---Q&A

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Playa Vista, California

The AeroSense Wavve, a remote contactless sleep monitor

Q1: Does the AeroSense Wavve support multi-user scenario?

A1: Unfortunately not right now. The AeroSense Wavve only supports a single-user-scenario. Any target within an 8ft (2.5m) range of the device may interfere with the accuracy of the detection and cause missing alerts or false alerts, for example, a nurse tending to the patient will generate a false “out-of-bed” alert.

Q2: How long does it take to get a reliable heart and breath rate reading?

A2: It's usually within 1 minute. If the user can be quiet and motionless, it will be faster.

Q3: Causes of fluctuation of the readings.

A3: It's usually caused by movements of the user monitored, for example when he turns on the bed or talks. Once such movement is picked up, it will take about another 30-60s for the readings to be stable again.

Q4: Why the Wavve cannot connect to the AeroSense Care APP?

A4: There are several possible causes:

1. The device is not connecting to the right WiFi network. Please reset the device and repeat the process.
2. The device is not yet assigned to any **Room** in the APP. Please assign the Wavve to **Bedroom**.
3. You may have used the wrong APP. The AeroSense Care APP is for the Wavve device. The AeroSense Hub APP is for our senior security and care kit.

Q5: The APP is not showing the real time heart and breath data correctly.

A5: It may take a few minutes to get the reading as the radar has to upload the data to the cloud first. However, it is also possible that the device is not installed correctly, and the radar is not facing the chest of the user. Please read the user manual carefully.

Q6: The LED indicator is red, and the device is not connected to the software.

A6: Please make sure to enter the current WiFi name and Password, as well as IP address. Please turn off the firewall of your computer. If you are using APP, it is possible the local WiFi connection is not stable enough, and it is recommended to use the personal hotspot of your cell phone.

Q7: The heart and breath data are not accurate.

A7: If the radar is installed correctly, the readings should be stable and accurate within 1 minute, given that the user is quiet and motionless. Otherwise, it will take longer than expected to get a reliable data.

The installation of the device may not be accurate, especially the height and the angle of the device. The user experience is the best if the chest is within 5ft (1.5m) away from the radar and the radar is facing directly to the chest.

Anyone else within a range of 8 ft (2.5m) may interfere with the result.

The results may not be that accurate if the user is covered by something very thick, lying on his side or stomach, or if the user is moving or talking.

The current algorithm supports a heart rate range of 55-90 bpm. Any heart rate beyond this range may not be accurately detected by the Wavve.

Q8: The Wavve shows readings even if there is nobody in bed.

A8: It can take up to 30s for the readings to return to zero once the user exits the bed.

It may be also caused by environmental noises, for example if the radar is not installed correctly and facing not toward the bed but other directions or the radar is facing at mirrors or large pieces of metals.

Q9: Software crashing or error messages.

A9: There are several possible causes.

1. Please end the process of "Aerosense tool" in the task manager and try to boot up the application again.
2. Please make sure there are 5 .dll files in the AeroSenseWavveTool file.
3. Your PC does not support Bluetooth.

Q10: The device can connected to the AeroSenseWavveTool or web-based software successfully, but cannot get any heart and breath curve.

A10: It may be blocked by your firewall. Please try the personal hotspot of your cell phone instead of your local WiFi network.

Q11: How to analyze the data?

A11: The data stream is in a bytes-array format. Please analyze the data using the communication protocol we provided. DO NOT transfer them to **string**.

Q12: The device communicates with the server via a TCP protocol. What is the Bluetooth for?

A12: The Wavve needs Bluetooth to setup the network, the WiFi and IP of the router. The device can then connect to the correct WiFi and send data to the correct IP address.

Q13: The device can be connected to the server, but it is not receiving any data, or the LED indicator is red.

A13: There are several possible causes.

1. The server is not setup correctly. Please refer to our sample codes.
2. If the server can receive data but the LED indicator is red: please make sure the server replies to the right protocol, and the LED indicator will be blue afterwards. Please refer to our communication protocol.
3. You've entered a wrong WiFi name or password. Please reset the device and enter the correct information.
4. Wrong IP/Port or the IP/Port has changed. Please reset the device and enter the correct information.
5. If your router has no password, please do not enter anything in password.
6. The device support 2.4GHz WiFi only, and English characters only. Please make sure to use the right network.
7. The device cannot connect to the WiFi. In the settings page of the router or your local network, check if there is a device with the name containing esp32c3. If not, please reset the Wavve and connect it to the personal hotspot in your cell phone to connect the Wavve to the network (2.4GHz and English characters only). Repeat the setup steps. Check the list of the devices that are connected to your personal hotspot and make sure the device with the name containing esp32c3 is connected.
8. There are connection issues between the current network and the server. Please use a PC to connect to the network and to test if the communication with the server is successful. If it fails, please turn on the local TCP server tool in your PC, reset the Wavve and configure it again, check if the local TCP server receive any data.
9. It may be blocked by firewalls. Please turn off all firewalls or other security software in your PC. Please also check the log of the server to make sure the port is not blocked.

The AeroSense Assure, a Home Care Assistant

Q1: The device fails to connect to the APP, and the LED indicators are red.

A1: There are several possible causes.

1. Wrong network name or password. Please reset the device and use the right name and password.
2. Please enter NONE in password if your router has no password.
3. Wrong type of network. The device supports 2.4GHz WiFi only, and there can only be English characters in the name of the network.
4. You may choose to connect the device WiFi when you should connect to the router WiFi. Please read the instructions carefully.
5. There are other devices that has been connected to your Assure. When you setup the Assure, it can be connected to only ONE cell phone or PC.

Q2: No alerts when fall occurs.

A: There are several causes.

1. The most common reason is de device is not installed correctly. Please read the manual carefully or contact our tech support for a DEMO video clip.
2. When the radar detects a potential fall, the LED indicators will flash red almost immediately. However, the alerts may be cancelled later if the radar detects vigorous movement of the target within the set buffer time (APP) or alert time (AeroSenseAssureTool), for example the user sits up or if someone else enters the FOV. In this case, you can see the LED indicators flash red but you will not get any alert messages or calls. (Please contact our tech support for a DEMO video clip.)
3. The FOV of the device is blocked. Please remove the block and try again.
4. When setup the parameters (buffer time or alert time, working distance and installation height), it doesn't match the actual figures. Please enter the correct value.
5. Please make sure there is no other moving objects or people within the working distance, 23ft (7m) by default.
6. Presence of mirrors or large pieces of metals (like a white board or a metal cabinet) within the FOV may interfere with the detection. Please do not install the radar directly opposite to mirrors or large pieces of metals.
7. If you still can not get any alerts, please contact our tech support.

Q3: When I test the fall accuracy, I can see the red flash, but I cannot get any calls or APP notifications.

A3: There are several possible causes.

1. The Axend AWS cloud is set so that there can be at most one call to the same number every hour. But you should be able to get the APP notification for all the falls.
2. When someone takes a fall but moves vigorously within the set buffer time or alert time (in which case the radar may consider the user having safely sit up or safely stood up), or if there is someone or other moving targets entering the FOV, the radar will cancel the alert.
3. If your Assure is using the firmware with learning algorithm, it is possible that your fall happens either within the silent learning period (the first 48 hours after activation), or the fall takes places where the learning algorithm categorized as false alerts, and therefore you are not getting any alerts.

Q4: About false alerts.

A4: The radar detects all kinds of moving targets. Possible causes are:

1. Interference caused by rotary fans or air conditioners. It is recommended to test the radar at least 6ft (2m) away from the interferences or to turn them off before testing. Our learning algorithm will be very effective to block these kinds of false alerts. Therefore, for everyday use, it is recommended to update your device to the latest firmware with learning algorithm.
2. Other common scenarios with false alerts including if the user is sitting or lying on a couch, sitting in front of a desk, or if the device is installed in an office or meeting room where there are a lot of people present. Whenever the radar detects the head moving followed by the feet moving, it is very like to generate false alerts. The learning algorithm can help to block some, but not all. Therefore, it is recommended to install the radar in rooms with simple layout such as bedroom or bathroom.
3. Pets or young kids can cause false alerts. Unfortunately, we still cannot find a better way around it without compromising the accuracy to detect falls.

Q5: When I update the firmware using the AeroSense Care APP, it constantly shows that the upgrade is in progress.

A5: The network is disconnected during update. Once the device is connected to the internet, the update will continue. Please wait and make sure the device is online.

Q6: The device is frequently offline.

A6: For APP users, please install the device where the WiFi signal is strong enough.

For developers, please set it at 2 min to decide whether the device is offline.

Q7: The device can connect to the AeroSense APP. But the server cannot receive any data when I use the AeroSenseAssure Tool to set up the device.

A7: If you can successfully setup the IP, it may be caused by your firewall which blocks the data.

Q8: If the user takes a fall and never stands up, will the Assure sends multiple alerts?

A8: No. The Assure will only send one alert. However, the LED indicators will keep flashing red unless the user gets up.

Q9: For developers with connection issues.

A9: There are several possible causes.

1. Please check the server settings to setup the environment and run the sample codes.
2. If the server can receive data but the LED indicator is red: please make sure the server replies to the right protocol, and the LED indicator will be blue afterwards. Please refer to our communication protocol.
3. It's possible that you've entered a wrong WiFi name or password. Please reset the device and enter the correct information.
4. Wrong IP/Port or the IP/Port has changed. Please reset the device and enter the correct information.
5. Please enter NONE in password if your router has no password.
6. The device support 2.4GHz WiFi only, and English characters only. Please make sure to use the right network.
7. The device cannot connect to the WiFi. In the settings page of the router or your local network, check if there is a device with the name containing USR-C210. If not, please reset the Wavve and connect it to the personal hotspot in your cell phone to connect the Wavve to the network (2.4GHz and English characters only). Repeat the setup steps. Check the list of the devices that are connected to your personal hotspot and make sure the device with the name containing USR-C210 is connected.
8. There are connection issues between the current network and the server. Please use a PC to connect to the network and to test if the communication with the server is successful. If it fails, please turn on the local TCP server tool in your PC, reset the Wavve and configure it again, check if the local TCP server receive any data.
9. It may be blocked by firewalls. Please turn off all firewalls or other security software in your PC. Please also check the log of the server to make sure the port is not blocked.

Support

Please contact support@ax-end.com for any help or more information.

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